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WHAT A CHANGE three months can make. Since Israel and the US bombed Iran on February 28, and the subsequent blockade of the Strait of Hormuz, global energy markets have gone into a tizzy.

For India, particularly vulnerable since 80% of its crude transits through the Hormuz, the war has triggered a fundamental rethink about the energy transition playbook.

76 | Coal is no longer a dirty fuel. Instead, as a transition fuel via coal gasification, it is now central to the country's plans to meet the growing industrial energy demand. After all, India holds one of the world's largest coal reserves, estimated at around 401 billion tonnes, along with nearly 47 billion tonnes of lignite reserves. Union Minister Ashwani Vaishnaw has said India's coal reserves were sufficient for nearly 200 years, making coal gasification an important pillar of long-term energy resilience.

Indeed, for India Inc, the shift to coal as a bridge fuel has been fast, especially where immediate electrification is difficult, as industrial operations have been hit by a shortage of Liquefied Natural Gas (LNG) amidst the West Asia conflict.

For instance, Jindal Steel recently switched to synthesis gas (syngas) through coal gasification due to a shortage of LNG supplies for its steel-making units. Ultra-

Tech is increasing the share of domestic coal in its energy mix in its cement plants. Mahindra & Mahindra is looking for electrification of fossil fuel processes across its ecosystem.

Boosting these efforts, the central government recently announced a ₹37,500 crore incentive scheme for coal gasification projects aimed at cutting India's ₹2.77 lakh crore import dependence on LNG, ammonia, methanol, fertilisers and other industrial feedstocks.

Of course, there is a trade-off here. India's carbon emissions will increase in a transition towards fossil-based power for energy security. This comes at a time when India has upped its climate action goals, pledging to achieve 60% of installed power capacity from non-fossil fuels by 2035. As part of its submission to the United Nations in March this year, the country committed to reducing emissions per unit of GDP by 47% by 2035 from 2005 levels.

Coal gasification, one of the solutions being touted now to offset the crude vulnerabilities, produces more carbon dioxide than a conventional coal-powered thermal power plant, according to an assessment by the think tank Centre for Science and Environment (CSE).

"It remains a carbon-intensive pathway. Without carbon capture, utilisation and storage (CCUS), coal gasification can lead to substantial CO emissions and lock in high-carbon infrastructure," says Parth Kumar, Programme Manager Sustainable Industrialisation Unit, CSE.

Of course, it isn't just coal that is being explored. There is a renewed focus on ramping up renewable energy capacity as well, along with a focus on nuclear

THE COAL REDUX

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Amid the current crisis, there is a renewed focus on increasing the usage of domestic coal by industries

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To bolster the economy, the government has announced a ₹37,500 crore incentive scheme for coal gasification

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However, the flip side of this increased reliance on coal is that carbon emissions are set to rise

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That has once again brought focus on carbon capture technology to offset the expected rise in emissions

